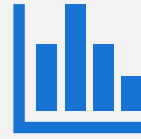

TCX2005: Information Systems, Management and Organisations

L7: Critical IS Infrastructure-II

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Lesson Objectives



Data and Information
Management



Telecommunications and
networking



How the Internet and wireless
technologies support business
operations

Learning Resources



- Textbook (read before class)
 - Read chapter 6, 7
- Any Other Material
 - Shared on Canvas for this week

Case Study

Read the case study from textbook on –[Astro: Leveraging Data for Customer-Driven Service](#):

Identify:

- a. The problem
- b. The solution
- c. Outcomes
- d. Key learning point(s)

Note: be ready to share your thoughts

Case Study

Problem:

- Local & international Competition (Maxis, Telekom Malaysia, iFlix, Netflix)
- Silos: data housed in multiple locations across various sites and cloud
- Legacy practices:
 - Structured data in centralized enterprise data warehouse
 - Unstructured data spread across multiple cloud-based systems
- Reduced visibility of data across the organization

Solution:

RM237-million investment in technology infrastructure

- Abandoned traditional enterprise data warehouse paradigm
 - Fully adopted cloud-based data lake approach
 - Built "Astro Data Lake" using Amazon Web Services (AWS)
 - Aggregated multiple data sources & touchpoints (views, transactions, customer interactions)
-

Case Study

Outcomes:

- Integrated previously siloed analytics across organizational divisions
- Introduced enterprise-wide best practices and standards on data quality
- Enhanced precise multi-channel advertising
- Gained in-depth understanding of customer media consumption
- Extended reach to 24 million people (2019)
- Launched new products and services including streaming platforms

Learning:

- Data as strategic asset for competitive advantage
 - Breaking down data silos improves overall data quality
 - Modern data architecture enables better integration
 - Cloud transformation provides scalability and advanced analytics
 - Integrating cross-channel data creates holistic customer insights
-

Case Study: Activity

- What are the various IT/IS tools & techniques mentioned in the case study?
- Also try to think what is a possible category for the tool/technique mentioned in the case such as- databases, networking, etc.

Case Study: Activity

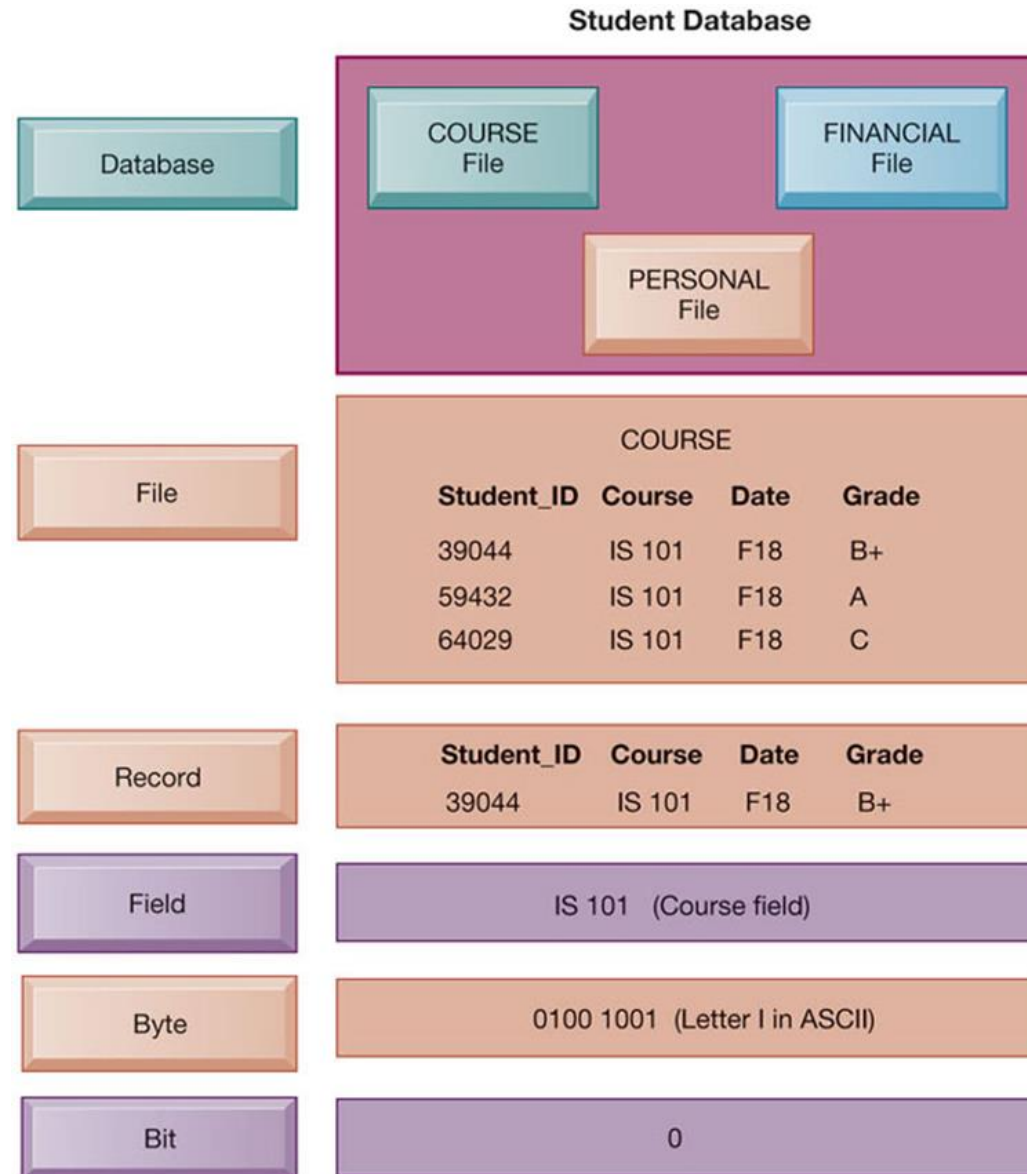
What are the various IT/IS tools & techniques mentioned in the case study?

[illegible]

File Organization Terms and Concepts

- **Database**: Group of related files
 - **File**: Group of records of same type
 - **Record**: Group of related fields
 - **Field**: Group of characters as word(s) or number(s)
 - **Entity**: Person, place, thing on which we store information
 - **Attribute**: Each characteristic, or quality, describing entity
-

Data Hierarchy



The Data Resource Challenge

Data has become a critical organizational asset

- Modern businesses generate and consume massive amounts of data

Problems with traditional file environments:

- Data redundancy and inconsistency
- Program-data dependence
- Lack of flexibility and poor security
- Limited data sharing and availability

Challenges of Traditional Data Environments

Poor Security

Varies in frequency but poses high risk; inconsistent controls increase vulnerability.



Limited Data Sharing

Common in siloed systems, leading to poor collaboration and fragmented decision-making.



Program-Data Dependence

Common in legacy systems; tight coupling between programs and data formats limits flexibility.



Data Redundancy

Frequently occurs and significantly impacts data accuracy, storage costs, and efficiency.



Example (Program-Data Dependence)

A company's customer file initially stores customer names in a 30-character field, and later needs to expand this to 50 characters to accommodate longer names, programmers would need to:

- Modify every program that accesses customer records
- Adjust all report formats
- Convert existing data
- Test all modified programs

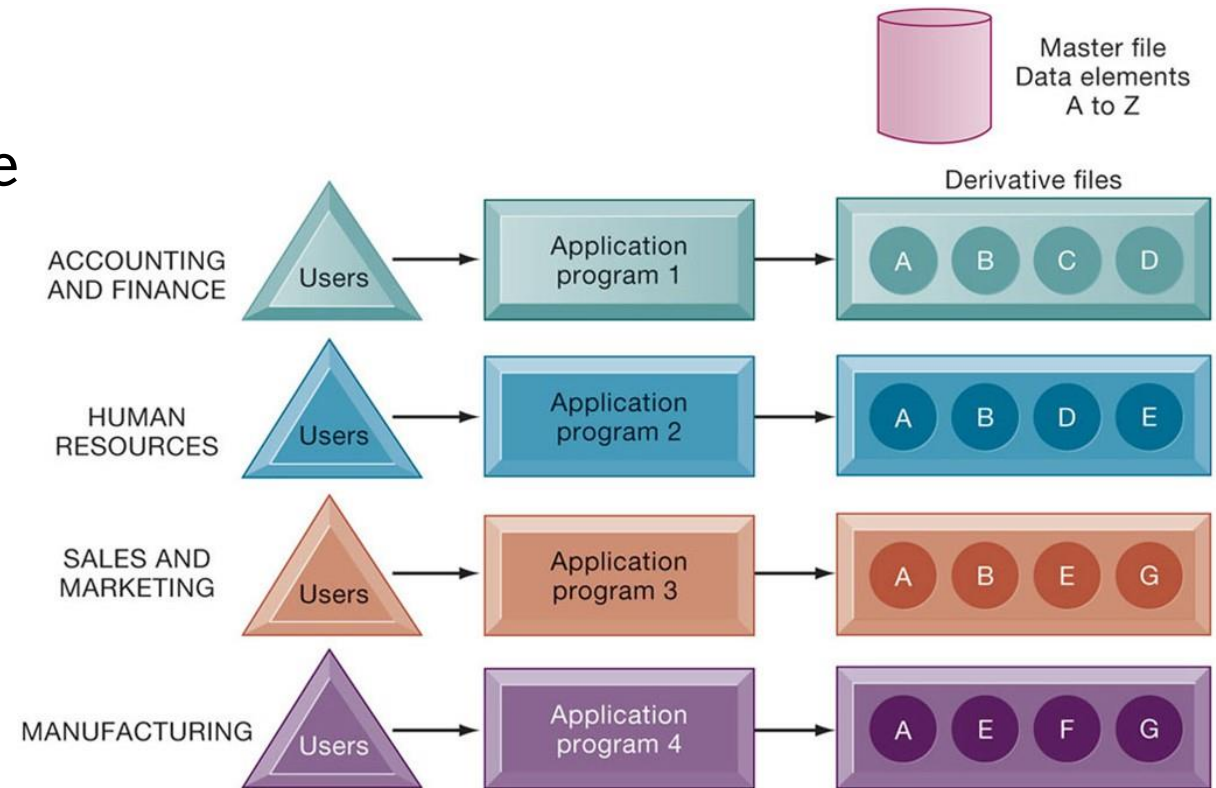
Solution: In modern DBMS, data definitions are stored separately from application programs.

- Programs access data through the DBMS interface
 - Data structures can change without modifying application code
 - Physical data independence is achieved (programs are isolated from how data is physically stored)
-

The Data Resource Challenge

Problems with traditional file processing:

- Each department maintains separate files
- Same data stored in multiple locations
- No coordination across business functions
- Difficult to generate integrated reports



Example

A customer moves to a new address.

In a traditional system, this change must be updated separately in the sales department's files, the billing department's files, and the shipping department's files.

If any department misses the update, inconsistencies arise.

Database Management Systems (DBMS)

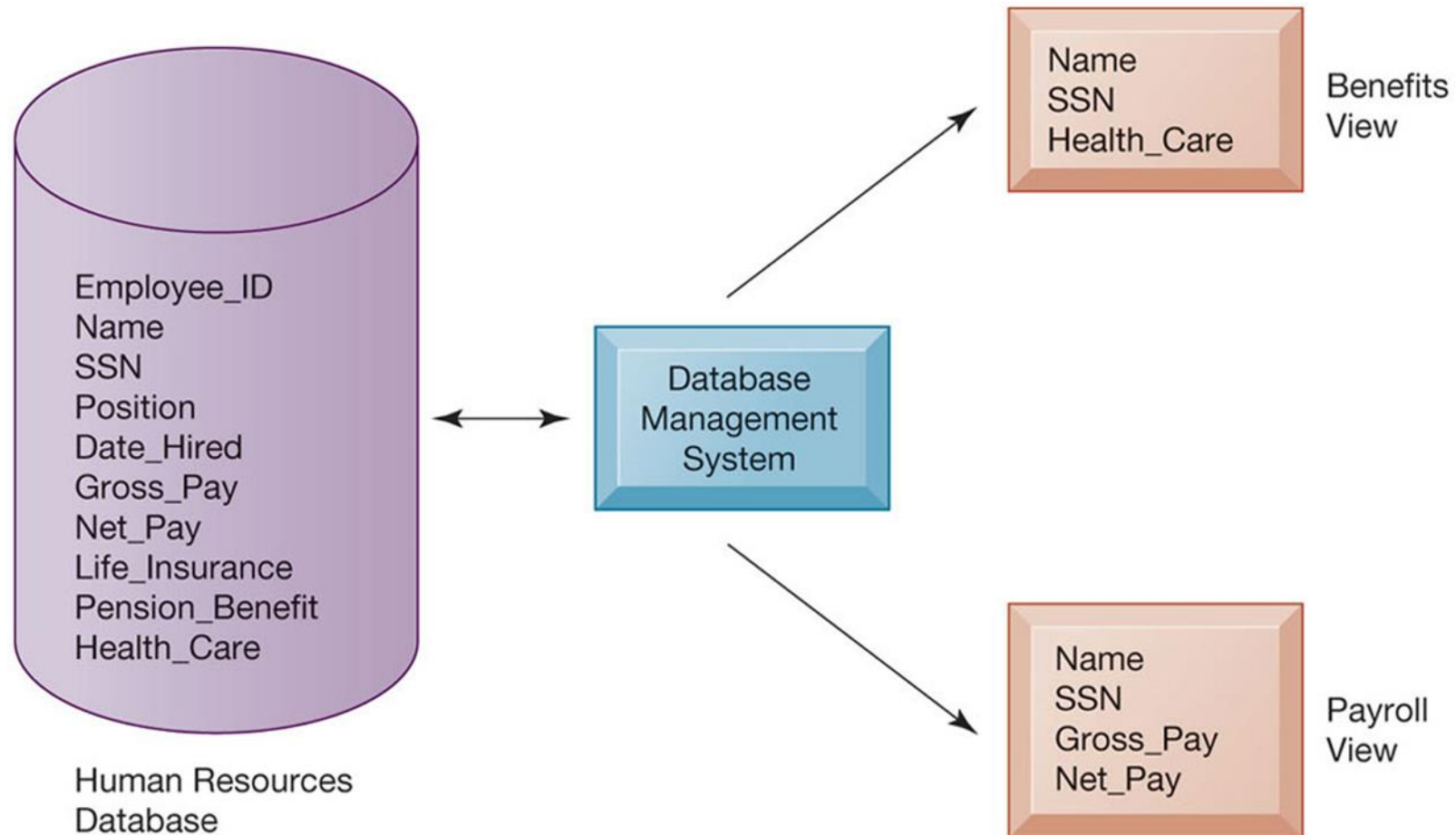
Database: Serves many applications by centralizing data and controlling redundant data

DBMS: Interfaces between applications and physical data files

- Provides consistent view of organizational data
- Solves traditional file environment problems:
 - Controls redundancy
 - Eliminates inconsistency
 - Separates programs and data
 - Enables centralized data management and security

Example: When a field like customer name is expanded from 30 to 50 characters, only the central data definition needs to be changed, not every program. Applications continue to work without modification because they access data through the DBMS interface rather than directly.

Example: DBMS



Human Resources Database with Multiple Views

Capabilities of Database Management Systems

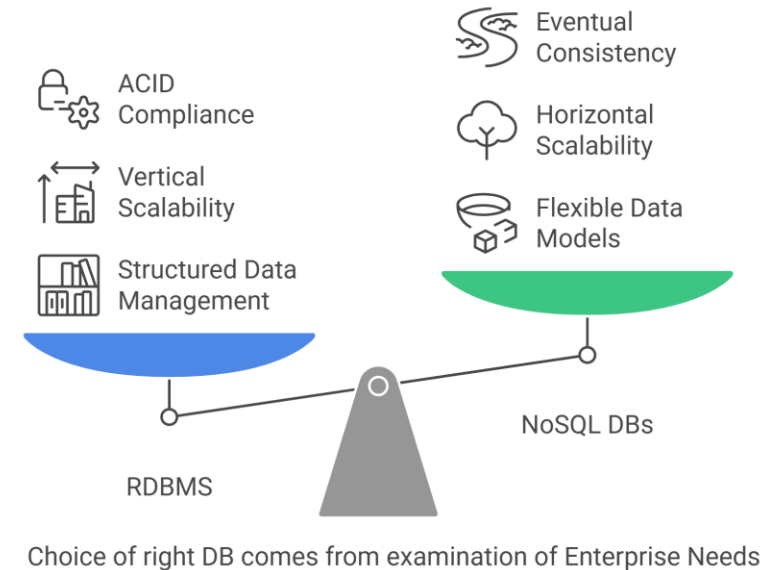
- Data definition capability
 - Data dictionary
- Querying and reporting
 - Data manipulation language
 - Structured Query Language (SQL)
- Many DBMS have report generation capabilities for creating polished reports (Microsoft Access)

Field Name	Data Type	Length	Description	Required
CustomerID	Integer	10	Unique customer identifier	Yes
FirstName	Varchar	50	Customer's first name	Yes
LastName	Varchar	50	Customer's last name	Yes
Email	Varchar	100	Customer's email address	Yes

Example: Data Dictionary

Non-Relational Databases and Databases in the Cloud

- **Non-relational databases: “No SQL”**
 - More flexible data model
 - Data sets stored across distributed machines
 - Easier to scale
 - Handle large volumes of unstructured and structured data



- **Databases in the cloud**
 - Appeal to start-ups, smaller businesses
 - Amazon Relational Database Service, Microsoft SQL Azure
 - Private clouds

The Challenge of Big Data

Big data: Massive sets of unstructured and semi-structured data

- Sources: web traffic, social media, sensors, IoT devices
- Volume too great for typical DBMS (petabytes, exabytes)
- Reveals patterns, relationships, and anomalies traditional systems miss
- Requires specialized tools and technologies

Example: A retail chain collects point-of-sale data (structured), customer reviews (semi-structured), security camera footage (unstructured), and in-store movement patterns from sensors (IoT data).

- Traditional databases struggle to manage this diverse data, but big data technologies can integrate these sources to reveal comprehensive customer behavior patterns.
-

Business Intelligence Infrastructure

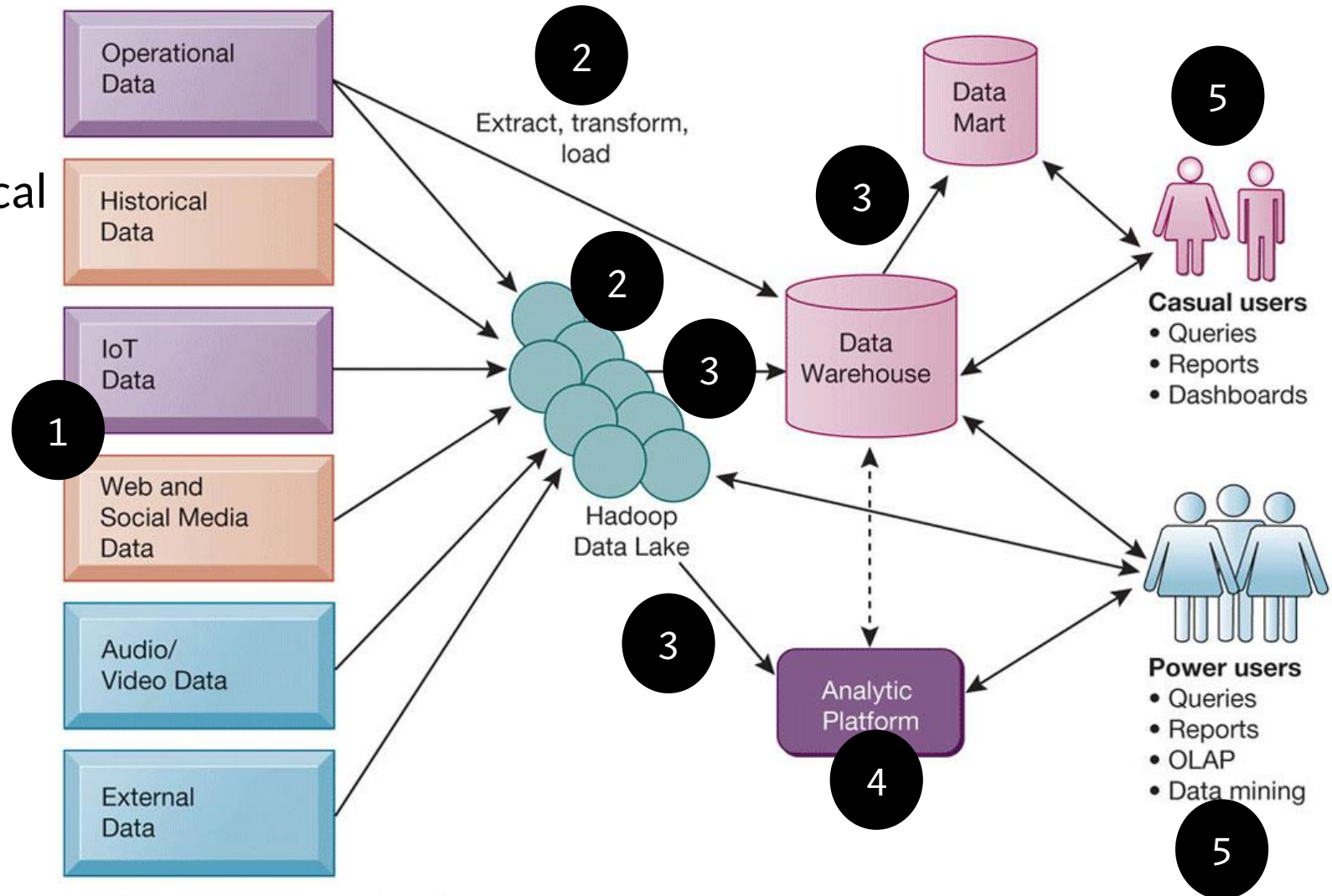
- Data warehouses:
 - Store current and historical data from operational systems
 - Stores current and historical data from many core operational transaction systems
 - Consolidates and standardizes information for use across enterprise, but data cannot be altered
 - Provides analysis and reporting tools
 - Data marts: Subject-specific subsets of data warehouses
 - Hadoop: Framework for distributed processing of big data
 - HDFS (Hadoop Distributed File System)
 - MapReduce for breaking data into clusters
-

Business Intelligence Infrastructure (cont.)

- Analytic platforms:
 - High-speed platforms using both relational and non-relational tools optimized for large datasets
 - In-memory computing
 - Used in big data analysis
 - Uses computers main memory (RAM) for data storage to avoid delays in retrieving data from disk storage
 - Can reduce hours/days of processing to seconds
 - Requires optimized hardware
-

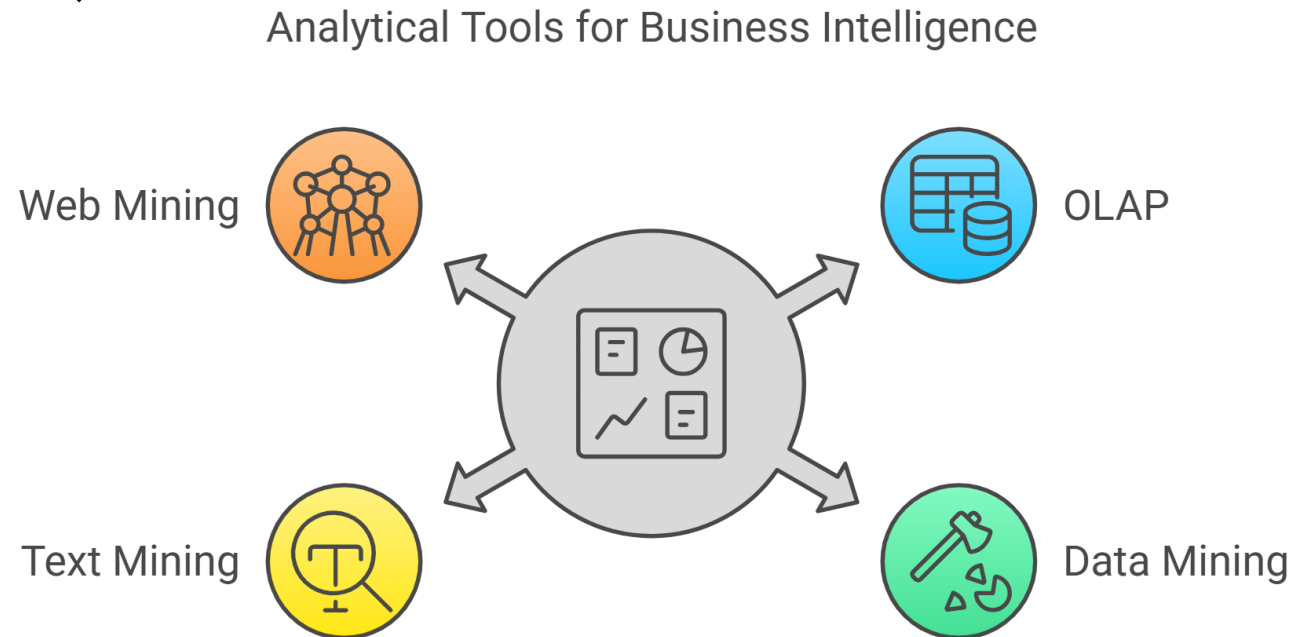
Example: Contemporary BI Infrastructure

- Data flows from operational systems to specialized analytical environments
- Multiple tools extract insights from structured and unstructured data



Analytical Tools for Business Intelligence

- Tools for consolidating, analyzing, and providing access to vast amounts of data to help users make better business decisions
 - Multidimensional data analysis (OLAP)
 - Data mining
 - Text mining
 - Web mining



Ensuring Data Quality and Governance

- More than 25 percent of critical data in Fortune 1000 company databases are inaccurate or incomplete
 - Before new database is in place, a firm must:
 - Identify and correct faulty data
 - Establish better routines for editing data once database in operation
-

Ensuring Data Quality and Governance (Cont.)

- **Data administration:** Establishes policies and procedures
 - **Information policy:** Rules for data sharing and management
 - **Data governance:** Manages data availability, integrity, and security
 - **Quality assurance:**
 - Data quality audits
 - Data cleansing
 - Monitoring and maintenance
-

Networking and Communication Trends

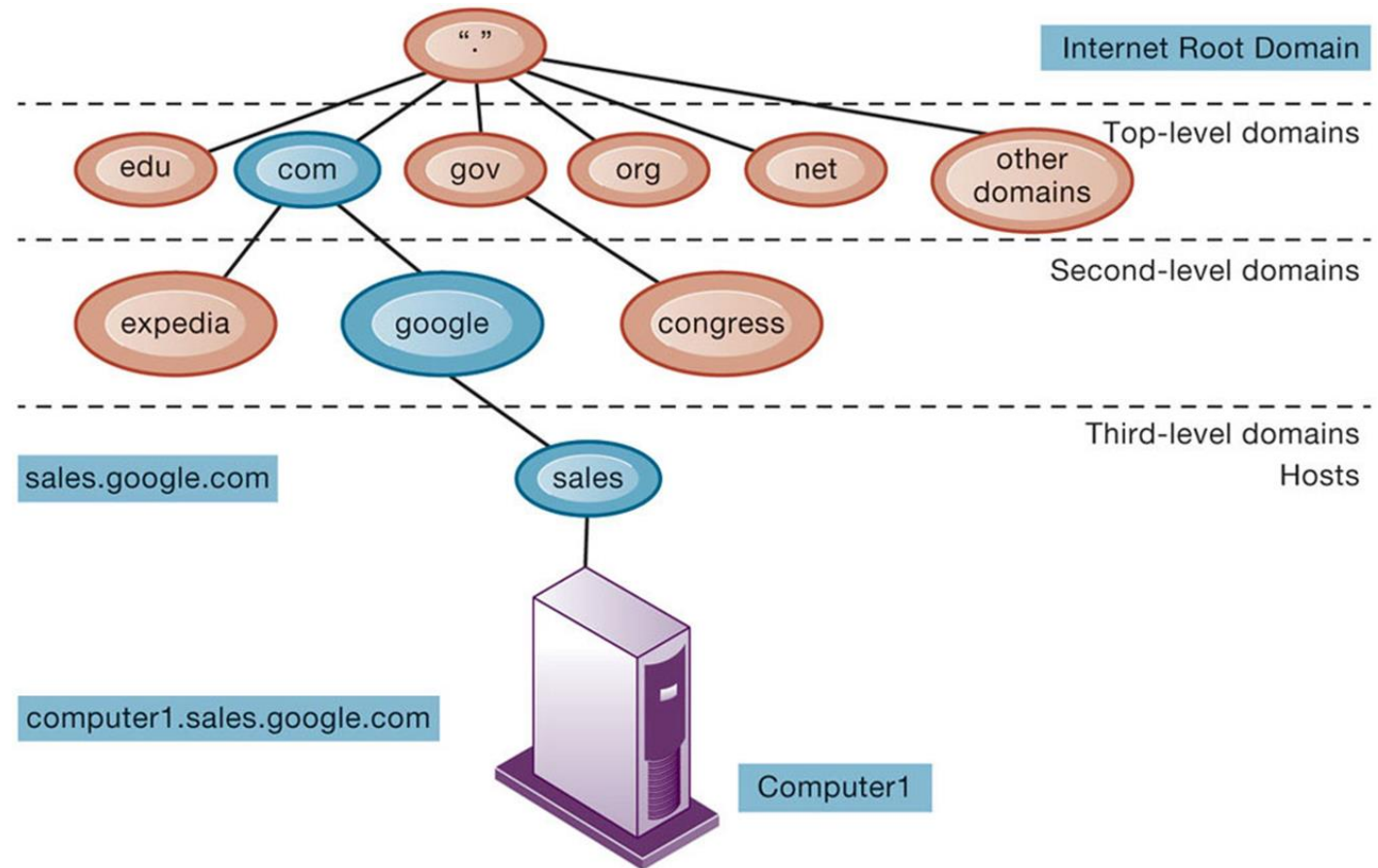
- **Convergence:** Telephone and computer networks merging.
E.g., Zoom, MS Teams
 - **Broadband:** High-speed internet access widespread
 - **Wireless technologies:** Enabling mobility and flexibility. E.g.,
Wifi, 5G
 - **Cloud-based networking:** Reducing infrastructure costs.
E.g., MS Azure, AWS
-

The Internet

- World's most extensive network
 - Internet service providers (ISPs)
 - Provide connections
 - Each device assigned an Internet Protocol (IP) address
 - Domain Name System (DNS) converts IP addresses to domain names
 - Network service providers own backbone networks
 - Governance bodies: IAB, ICANN, W3C
-

The Internet Architecture

- Hierarchical structure of networks
- Regional and global interconnections
- No single controlling entity



Wireless Technologies

- Cellular systems
 - 3G, 4G, 5G networks
 - Increasing speeds and capabilities
 - **Example:**
 - 4G networks typically offer speeds of 20-100 Mbps, sufficient for video streaming and basic business applications.
 - 5G networks can reach 1-10 Gbps, enabling real-time remote healthcare monitoring and augmented reality (AR) applications.
-

Wireless Technologies (cont.)

- Wi-Fi (802.11)
 - It is based on the IEEE 802.11 family of standards,
 - Wireless LANs
 - Hotspots for public access
 - **Example:**
 - A university campus deploys multiple Wi-Fi access points throughout buildings, allowing students and faculty to connect to the network from classrooms, libraries, and common areas without physical cables.
-

Wireless Technologies (cont.)

- Bluetooth (802.15)
 - Links up to 8 devices in 10-meter area using low-power, radio-based communication
 - Useful for personal networking (PANs)
 - **Example:**
 - In a modern conference room, presenters can connect their laptops wirelessly to displays, speakers can connect to microphone systems, and participants can share files directly between devices using Bluetooth connections.
-

Wireless Technologies (cont.)

- Others
 - Cellular Systems
 - IoT devices and Sensors
 - Wireless LAN (WLAN)
 - RFID and Wireless Sensor Networks (WSNs)
-

Implications for Business

- **Operational efficiency:** Streamlined processes and workflows
 - **Decision support:** Data-driven insights for management
 - **Customer service:** Enhanced responsiveness and personalization
 - **Strategic advantage:** New business models and market opportunities
 - **Mobility:** Work from anywhere, anytime
 - **Challenges:** Security, privacy, complexity
-

Example: Enterprise Networks in Action

- Smart Retail Experience: [Alibaba's Hema app](#)
 - Integration of online and offline shopping
 - Leverages:
 - Mobile app integration
 - QR code scanning
 - Data analytics
 - IoT sensors
 - Real-time inventory management
-

Example: Enterprise Networks in Action (cont.)

- The Hema app continuously acquires information from its customers
 - Demonstrates IT's role in helping organizations increase efficiency and lower costs
 - Illustrates the ability of IT systems to support inventory management and sales
-

Future Trends

- 5G and beyond
 - Edge computing
 - Zero-trust network architecture
 - Increased AI integration
 - Quantum technologies
 - Expanded IoT ecosystems
 - Blockchain for secure transactions
 - Etc....
-

Next

Securing Information Systems:
Governance & Risk Management
